

Unit 7

Basis and dimension.

Unit Goals

- Be able to explain what **basis of a vector space** is and how it relates to dimension.
- Be able find a **basis for the row space** of a matrix using forward elimination.
- Be able find a **basis for the column space** of a matrix using the casting out algorithm.
- Be able find a **basis for the null space** of a matrix by computing the special solutions.
- Be able to explain the relationship between the four fundamental subspaces of a matrix.

Unit 7.1

**Linear independence,
dimension and basis.**

Linear dependence and independence.

- If at least one vector in a set can be written as a linear combination of the others, we say that the set is **linearly dependent**.
- If no vectors from the set can be written as linear combinations of the others, we say the set is **linearly independent**.

$R = \{(1,0,1), (1,0,-1)\}$ is linearly independent.

$S = \{(1,0,1), (1,0,-1), (2,0,2)\}$ is linearly dependent.

$$(2,0,2) = 2(1,0,1)$$

$T = \{(1,0,1), (1,0,-1), (3,0,1)\}$ is linearly dependent.

$$(3,0,1) = 2(1,0,1) + (1,0,-1)$$

Linear dependence and independence.

- As the rows and columns of a matrix can be considered set of vectors, we can discuss their linear dependence/independence.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 2 & 2 & 4 \\ 0 & 0 & 3 & 3 \end{bmatrix}$$

- The row and columns of A are **both linearly dependent**.
- The row and columns of B are **both linearly independent**.
- The **rows** of C are **linearly independent**, but the **columns** of C are **linearly dependent**.

Dimension of a linear span.

- The **linear span** of a set of vectors is the set of all possible linear combinations of those vectors; we call the set of vectors we begin with a **spanning set**.
- If a spanning set is linearly dependent, we can remove any linearly dependent vectors without changing the span; the linearly dependent vectors can be “reconstructed” by linear combinations:

$$\begin{aligned}R &= \{(1,0,1), (1,0,-1)\} \\S &= \{(1,0,1), (1,0,-1), (2,0,2)\} \\T &= \{(1,0,1), (1,0,-1), (3,0,1)\}\end{aligned}$$

$$U = \text{span}(R) = \text{span}(S) = \text{span}(T) = \{\vec{x} : \vec{x} = s(1,0,1) + t(1,0,-1), \forall s, t \in \mathbb{R}\}$$

- The **dimension** of a linear span is the number of *linearly independent* vectors in a spanning set:

$$\dim(U) = \dim(\text{span}(R)) = 2$$

Geometric meaning of dimension.

Definitions:

- The **dimension** of any vector space is the number of *linearly independent* vectors in a spanning set for the space.
- The dimension tells us how many vectors we need to generate the space.

Geometric interpretation:

- The dimension of a vector space defines its geometry:
 - A vector space of dimension 0 is a **point** (no generating vectors $\rightarrow$ origin only).
 - A vector space of dimension 1 is a **line** (generated by one vector).
 - A vector space of dimension 2 is a **plane** (generated by two vectors).

⋮

Basis of a vector space.

- The **dimension** of any vector space is the number of **linearly independent vectors** in a spanning set for the space.
- If a spanning set has **more vectors than the dimension**, some must be linearly dependent (redundant)
- If a set has **fewer vectors than the dimension**, it cannot be a spanning set. For e.g., a plane cannot be defined by a vector.
- Any linearly independent spanning set for a vector space is called a **basis** for that vector space.

Basis and dimension (Recap)

- A **basis** for a vector space:
 - Must be linearly independent $\rightarrow$ no more vectors than necessary.
 - Must be a spanning set $\rightarrow$ no fewer vectors than necessary.
- The **dimension** of a vector space:
 - Is the number of vectors in any basis.
 - Defines the geometry of the space.
 - Is a single number (constant for a given vector space).

Basis and dimension.

Outstanding questions:

- Given a set of vectors that span a vector space, how do we find the dimension and a basis?
- Given a set of vectors, how do we know if they are linearly independent?

The row space of a matrix.

- The row space of a matrix is the **linear span** of the vectors corresponding to each row.
- We define a spanning set from the rows:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} (\vec{a}_1^R)^T \\ (\vec{a}_2^R)^T \\ \vdots \\ (\vec{a}_m^R)^T \end{bmatrix}, (\vec{a}_i^R)^T = [a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}]$$

$$S_R = \{\vec{a}_1^R, \vec{a}_2^R, \dots, \vec{a}_m^R\}$$

- and this gives us the row space:

$$R(A) = \text{rowsp}(A) = \text{span}(S_R) = \text{span}(\{\vec{a}_1^R, \vec{a}_2^R, \dots, \vec{a}_m^R\})$$

Linear dependence and elimination.

- During elimination, we take linear combination of rows.
- If a set of vectors can be linearly combined using non-zero coefficients to give the zero vector, they must be **linearly dependent**:

$$a\vec{u} + b\vec{v} + c\vec{w} = 0 \leftrightarrow \vec{w} = -\frac{a}{c}\vec{u} - \frac{b}{c}\vec{v}, c \neq 0$$

- **Zero rows will result during elimination if (and only if) the rows were linearly dependent.**
- By definition any linear combination of vectors from a vector space must also be in the vector space.
- **Every matrix generated during the process of elimination has the same row space:** $A \rightarrow U \rightarrow R \Leftrightarrow R(A) = R(U) = R(R)$

Row space: Basis and dimension.

- The non-zero rows of any echelon form of a matrix are a basis for its row space.
- The number of non-zero rows of any echelon form of a matrix is the dimension of the row space.

The Row Space algorithm to find a basis from a spanning set:

- Create a matrix whose rows are the spanning set.
- Carry out elimination to either U or R .
- The resulting non-zero rows are a basis.

$$T = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m\} \rightarrow A = \begin{bmatrix} \vec{v}_1^T \\ \vec{v}_2^T \\ \vdots \\ \vec{v}_r^T \\ \vec{v}_{r+1}^T \\ \vdots \\ \vec{v}_m^T \end{bmatrix} \rightarrow \begin{bmatrix} \vec{u}_1^T \\ \vec{u}_2^T \\ \vdots \\ \vec{u}_r^T \\ \vec{0} \\ \vdots \\ \vec{0} \end{bmatrix} = U \text{ or } R \rightarrow \begin{array}{l} \text{Basis: } S = \{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_r\} \\ \dim(\text{span}(S)) = r \\ \text{span}(T) = \text{span}(S) \end{array}$$

Row space: Basis and dimension.

- Example – Basis and dimension of the row space of a matrix.

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 2 & 3 & 3 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$$\dim(\mathbf{R}(A)) = 3, \quad \text{Basis of } \mathbf{R}(A): \{(1,1,1,1), (0,1,1,1), (0,0,0,1)\}$$

Row space: Basis and dimension.

- Example – Basis and dimension using the row space algorithm:

$$V = \text{span}(S), S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \right\}$$

$$S \rightarrow A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 0 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -3 \\ 0 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

$$\dim(V) = \dim(\mathbf{R}(A)) = 2, \quad \text{Basis of } V: \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

The column picture of a linear system.

- $\vec{x}$ represents the coefficients needed to express $\vec{b}$ as a linear combination of the columns of A :

$$A\vec{x} = \vec{b} \rightarrow [\vec{c}_1 \quad \vec{c}_2 \quad \cdots \quad \vec{c}_n]\vec{x} = \vec{b} \rightarrow x_1\vec{c}_1 + x_2\vec{c}_2 + \cdots + x_n\vec{c}_n = \vec{b}$$

- The column space of A is the linear span of the column:.

$$C(A) = \text{span}(\{\vec{c}_1, \vec{c}_2, \dots, \vec{c}_n\})$$

- There will be **a solution** if and only if $\vec{b}$ is in the column space of A .
- There will be **a unique solution** only if the columns of A are **linearly independent**.
- The columns of A are **linearly independent** if and only if the null space of A is the zero vector: there are no free variables; every column contains a pivot.

Elimination and the column space.

- Elimination takes linear combination of rows.
- The column space is not preserved during elimination:
$$A \rightarrow U \rightarrow R \Leftrightarrow \mathcal{C}(A) \neq \mathcal{C}(U), \mathcal{C}(A) \neq \mathcal{C}(R)$$
- Elimination is driven by pivots; columns with free variables have no impact on the elimination steps $\rightarrow$ we can remove them!
- Removal of a free variable column cannot introduce a zero row, thus cannot change the existence of a solution $\rightarrow$ column space is preserved.
- Removal of a free variable column reduces the number of solutions; a unique solution exists only if there are no free variables $\rightarrow$ linear independent columns.

Elimination and the column space.

• **Example:** $A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & -1 & 3 \\ 1 & 2 & 3 & -2 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 10 \\ -2 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 & 1 & 0 & 2 \\ 2 & 4 & -1 & 3 & 10 \\ 1 & 2 & 3 & -2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 & 2 \\ 0 & 0 & -3 & 3 & 6 \\ 0 & 0 & 2 & -2 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 & 2 \\ 0 & 0 & -3 & 3 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 & 4 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_p = \begin{bmatrix} 4 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \vec{x}_{n1} = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 10 \\ 1 & 3 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & -3 & 6 \\ 0 & 2 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & -3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_p = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

Column space: Basis and dimension.

- The number of pivots in any echelon form of a matrix is the dimension of the column space.
- The **pivot columns of the original matrix are basis for its column space.**

The Casting-Out algorithm to find a basis from a spanning set:

1. Create a **matrix** whose columns are the spanning set.
2. Carry out **elimination** to either U or R .
3. The pivot columns from the original matrix are a **basis**.

Column space: Basis and dimension.

- Example – Basis and dimension of the column space of a matrix.

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 2 & 3 & 3 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$$\dim(\mathcal{C}(A)) = 3, \quad \text{Basis of } \mathcal{C}(A): \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 4 \end{bmatrix} \right\}$$

The casting-out algorithm

- Example – Basis and dimension using the casting-out algorithm:

$$V = \text{span}(S), S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \right\}$$

$$S \rightarrow A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 1 & -1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 2 \\ 0 & -3 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix} = U$$

$$\dim(V) = \dim(\mathbf{C}(A)) = 2, \quad \text{Basis of } V: \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} \right\}$$

Unit 7.2

Basis, dimension and the relationships of the four subspaces of a matrix.

Row space, column space and rank.

- Forward elimination gives the dimension of the row space of a matrix by counting the number of non-zero rows in a row echelon form.
- Forward elimination gives the dimension of the column space by counting the number of pivots in a row echelon form.
- A pivots are just the first non-zero entry in a row; the number of non-zero rows and the number of pivots are equal:
- **The dimension of the row space and the column space of a matrix are the same; this is the rank of the matrix:**

$$\text{rank}(A) = \dim(\mathbf{R}(A)) = \dim(\mathbf{C}(A))$$

Parent spaces of an m -by- n matrix.

- Each row of an m -by- n matrix is a vector with n elements.
- The parent space of the row space is the n -D space.

$$A \in \mathbb{R}^{m \times n} \rightarrow \mathbf{R}(A) \subseteq \mathbb{R}^n$$

- Each column of an m -by- n matrix is a vector with m elements.
- The parent space of the column space is the m -D space.

$$A \in \mathbb{R}^{m \times n} \rightarrow \mathbf{C}(A) \subseteq \mathbb{R}^m$$

- Note on notation: $\subseteq$ reads as “is a subset of” or “is a subspace of”

The null space of a matrix.

- The set of all possible solutions to the homogenous system, $A\vec{x} = \vec{0}$, is called the null space of A .

$$N(A) = \text{nullsp}(A) = \left\{ \vec{x} : A\vec{x} = \vec{0} \right\}$$

- The null space is a vector space:

$$\vec{u}, \vec{v} \in N(A) \rightarrow k\vec{u} + l\vec{v} \in N(A), \forall k, l \in \mathbb{R}$$

- We can describe the null space by a set of special solutions, one per free variable:

$$\{ \vec{x} : \vec{x} = s_1\vec{x}_{n1} + s_2\vec{x}_{n2} + \cdots + s_l\vec{x}_{nl}, \forall s_i \in \mathbb{R} \}$$

Null space: Basis and dimension.

- The special solutions are created to be linearly independent:
 - Because a different free variable is non-zero in each special solution, we can not create one special solution from the others.
- The special solutions are sufficient to describe all solutions:
 - The general solution is the set of linear combinations of the special solutions.
- The special solutions are a basis for the null space!
- The number of free variables in an echelon form is the dimension of the null space.

Relationship of null and row spaces.

- The parent space of the row space is the n -D space.

$$A \in \mathbb{R}^{m \times n} \rightarrow \mathbf{R}(A) \subseteq \mathbb{R}^n$$

- The dimension of the row space is the rank (the number of pivots):

$$\dim(\mathbf{R}(A)) = \text{rank}(A)$$

- The parent space of the null space is also the n -D space.

$$A\vec{x} = \vec{0}, A \in \mathbb{R}^{m \times n} \rightarrow \vec{x} \in \mathbb{R}^n \rightarrow \mathbf{N}(A) \subseteq \mathbb{R}^n$$

- The dimension of the null space is the number of free variables:

$$\dim(\mathbf{N}(A)) = n - \dim(\mathbf{R}(A)) = n - \text{rank}(A)$$

Relationship of null and row spaces.

- The row and null space share the same parent space:

$$A \in \mathbb{R}^{m \times n} \rightarrow \mathbf{R}(A) \subseteq \mathbb{R}^n, \mathbf{N}(A) \subseteq \mathbb{R}^n$$

- The sum of the dimensions of the row and null spaces equals the dimension of the parent space:

$$\dim(\mathbf{R}(A)) + \dim(\mathbf{N}(A)) = \dim(\mathbb{R}^n), A \in \mathbb{R}^{m \times n}$$

- The dimension of the row space is the rank; we give a special name to the dimension of the null space as well — the nullity:

$$\text{nullity}(A) \equiv \dim(\mathbf{N}(A))$$

$$\text{rank}(A) + \text{nullity}(A) = n, A \in \mathbb{R}^{m \times n}$$

The null and row spaces are linearly independent.

- For all vectors in the null space: $A\vec{x} = \vec{0}$.
- For each $\vec{r}_i \neq \vec{0}$:

$$A\vec{r}_i = \begin{bmatrix} \vdots \\ \vec{r}_i^T \\ \vdots \end{bmatrix} \vec{r}_i = \begin{bmatrix} \vdots \\ \vec{r}_i^T \vec{r}_i \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ \|\vec{r}_i\|^2 \\ \vdots \end{bmatrix} \neq \vec{0}$$

- The non-zero rows cannot be in the null space $\rightarrow$ non-zero linear combinations of these rows cannot be in the null space; the only vector that is in both the row and null space is the zero vector.
- The union of a basis for the null space and a basis of the row space contains n linearly independent vectors $\rightarrow$ a basis for $\mathbb{R}^n$.

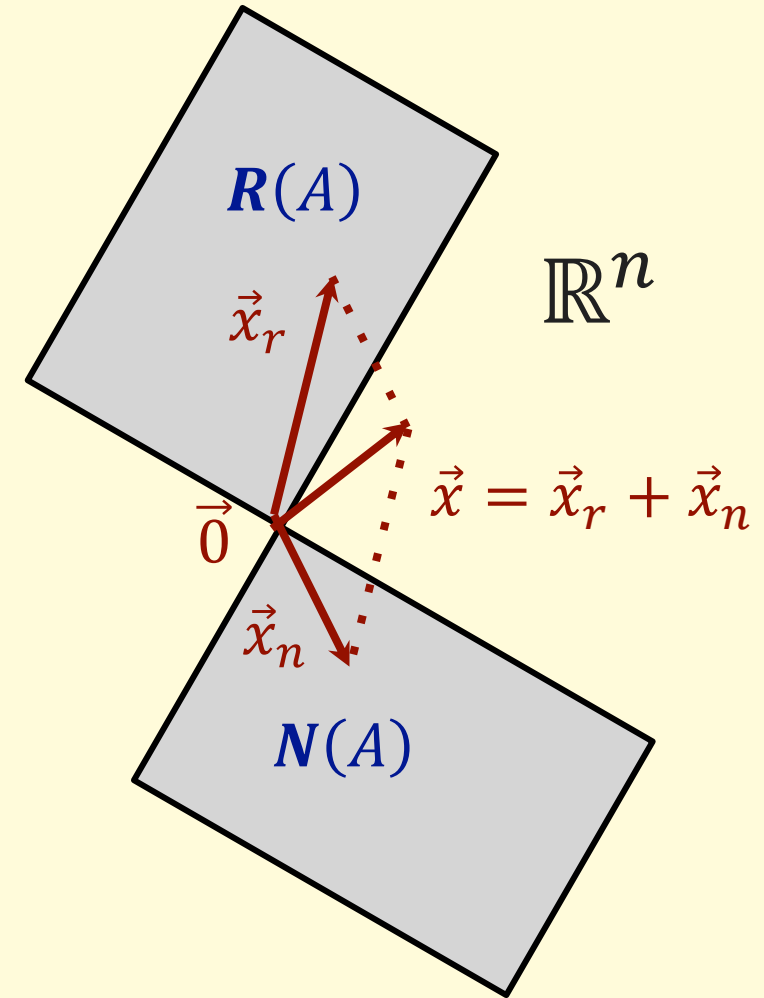
$\mathbb{R}^n$ is the direct sum of the row and null spaces.

- A direct sum of two vector space is the space of all linear combinations of vectors from the two spaces:

$$R(A) \oplus N(A) = \mathbb{R}^n$$

- Every vector in $\mathbb{R}^n$ can be expressed as unique sum of vectors from the null space and row space:

$$\begin{aligned} &\forall \vec{x} \in \mathbb{R}^n, \\ &\exists! \vec{x}_r \in R(A), \vec{x}_n \in N(A): \\ &\quad \vec{x} = \vec{x}_r + \vec{x}_n \end{aligned}$$



Row space, null space and the general solution to a system.

- We can write the general solution for $A\vec{x} = \vec{b}$ as the sum of a particular solution and a linear combination of the special solutions to $A\vec{x} = \vec{0}$:

$$\vec{x} = \vec{x}_p + s_1\vec{x}_{n1} + s_2\vec{x}_{n2} + \cdots + s_{(n-r)}\vec{x}_{n,(n-r)}$$

- Every vector in $\mathbb{R}^n$ can be expressed as unique sum of vectors from the null space and row space:

$$\forall \vec{x} \in \mathbb{R}^n, \exists! \vec{x}_r \in R(A), \vec{x}_n \in N(A): \vec{x} = \vec{x}_r + \vec{x}_n$$

- Every solution to $A\vec{x} = \vec{b}$ has a *unique* particular solution that is in the row space.

- **Note, that this is not generally the solution we get by setting all free variables equal to zero! (We can find this solution using the SVD – later in the semester).**

Row and null spaces of A^T

- The column space of any matrix is the row space of its transpose.
- The row space of any matrix is the column space of its transpose.

$$\mathbf{C}(A) = \mathbf{R}(A^T), \quad \mathbf{R}(A) = \mathbf{C}(A^T)$$

- The row and null spaces are subspaces of $\mathbb{R}^n$:

$$A \in \mathbb{R}^{m \times n} \rightarrow \mathbf{R}(A) \subseteq \mathbb{R}^n, \mathbf{N}(A) \subseteq \mathbb{R}^n, \quad \mathbf{R}(A) \oplus \mathbf{N}(A) = \mathbb{R}^n$$

- The column space is a subspace of $\mathbb{R}^m$:

$$A \in \mathbb{R}^{m \times n} \rightarrow \mathbf{C}(A) \subseteq \mathbb{R}^m$$

- Define a left null space, the null space of A^T :

$$A \in \mathbb{R}^{m \times n} \rightarrow A^T \in \mathbb{R}^{n \times m} \rightarrow \mathbf{N}(A^T) \subseteq \mathbb{R}^m$$

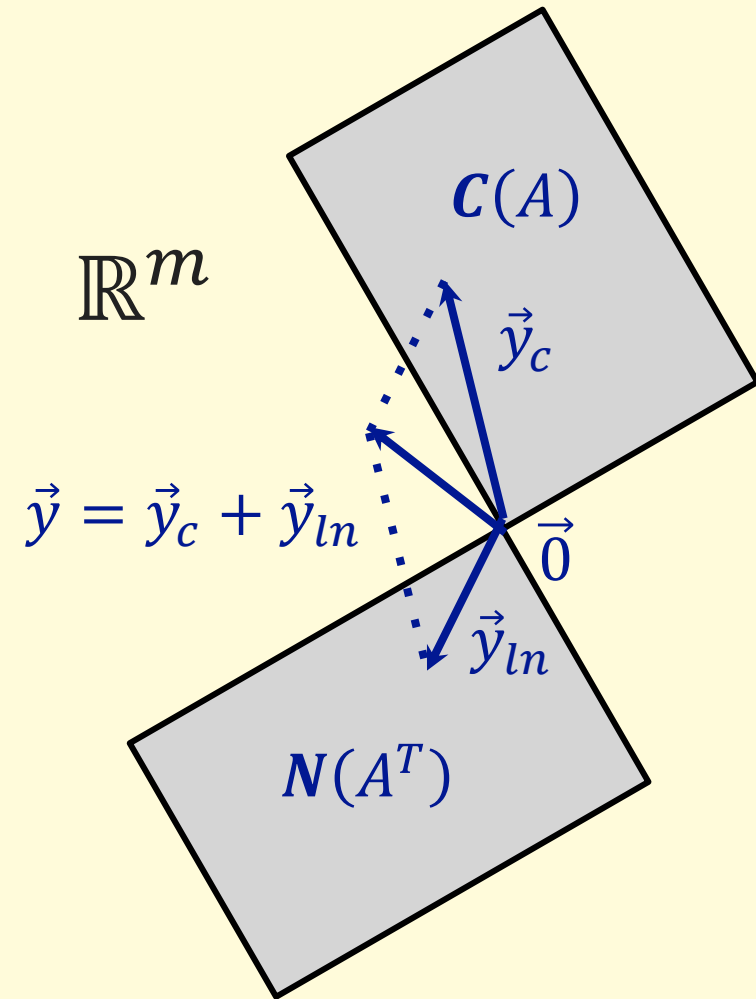
$\mathbb{R}^m$ is the direct sum of the column and left null spaces.

$$A \in \mathbb{R}^{m \times n} \rightarrow A^T \in \mathbb{R}^{n \times m}$$
$$\mathbf{R}(A^T), \mathbf{N}(A^T) \subseteq \mathbb{R}^m$$

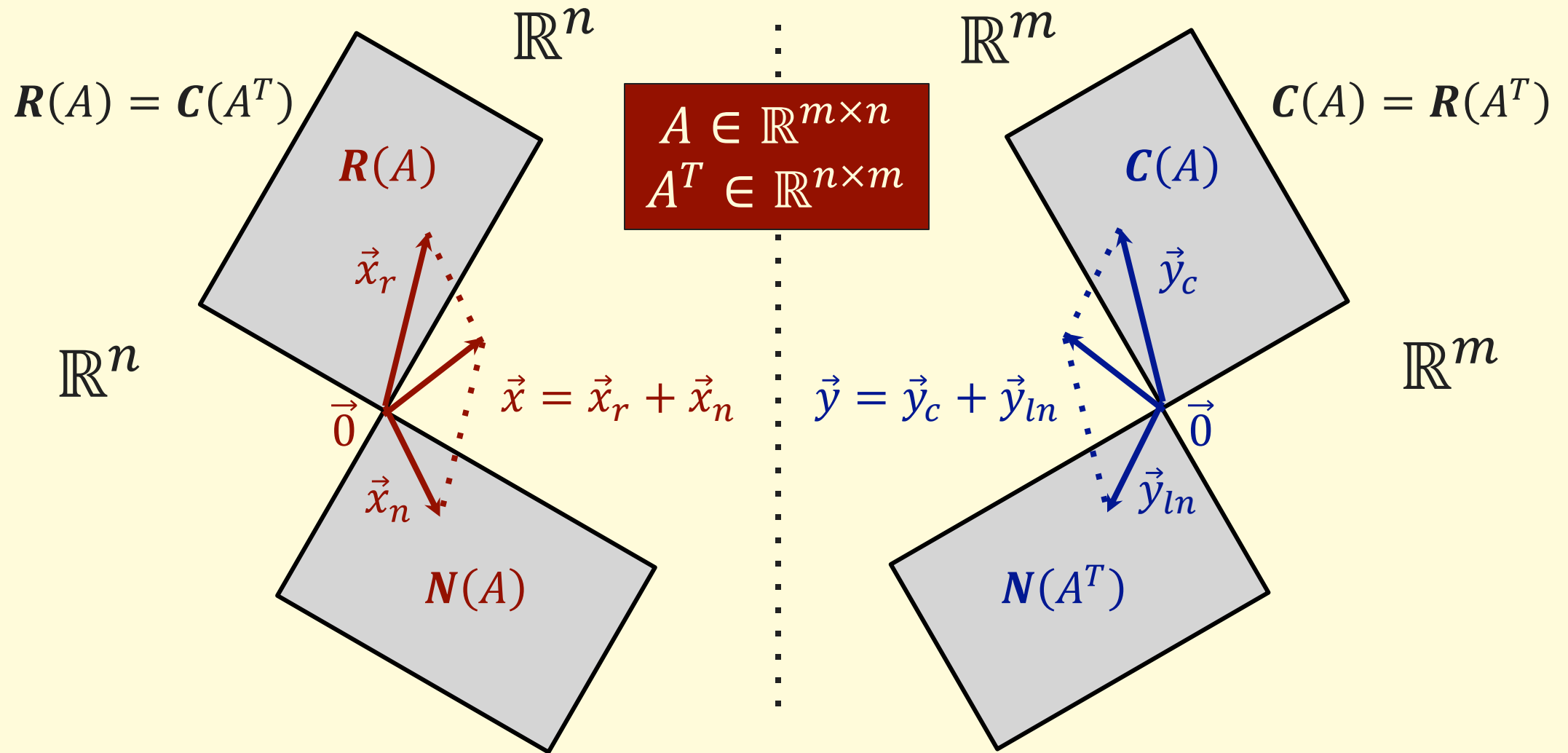
$$\mathbf{R}(A^T) \oplus \mathbf{N}(A^T) = \mathbb{R}^m$$
$$\mathbf{C}(A) = \mathbf{R}(A^T)$$

- Every vector in $\mathbb{R}^m$ can be expressed as unique sum of vectors from the column space and the left null space:

$$\forall \vec{y} \in \mathbb{R}^m,$$
$$\exists! \vec{y}_c \in \mathbf{C}(A), \vec{y}_{ln} \in \mathbf{N}(A^T):$$
$$\vec{y} = \vec{y}_c + \vec{y}_{ln}$$



The fundamental matrix subspaces.



One matrix, two systems.

- For any matrix, we can define two linear systems, using either the original matrix or its transpose as the coefficient matrix.

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{m \times n}, \vec{x} \in \mathbb{R}^n, \vec{b} \in \mathbb{R}^m$$

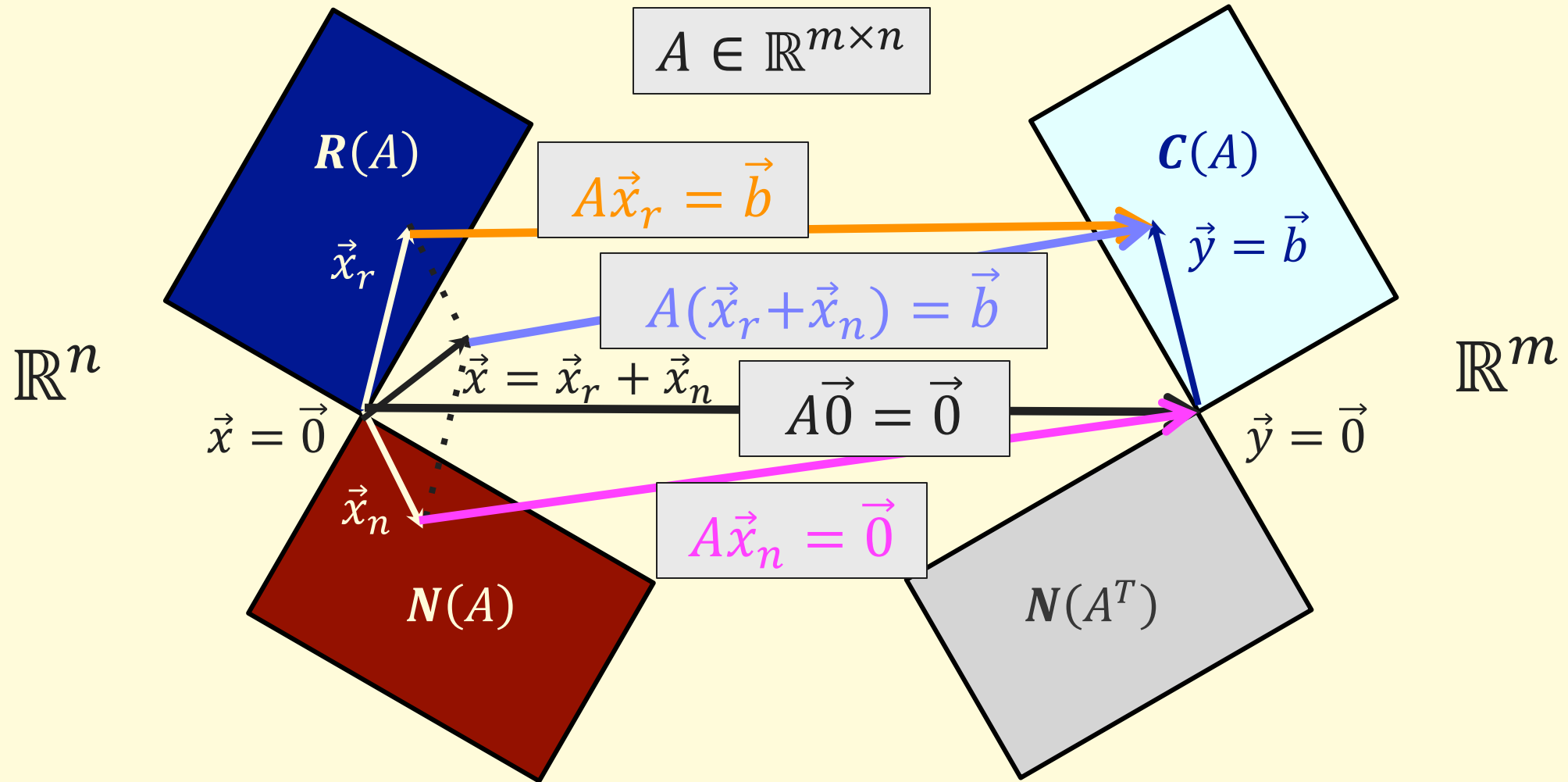
$$A^T \vec{y} = \vec{c}, A^T \in \mathbb{R}^{n \times m}, \vec{y} \in \mathbb{R}^m, \vec{c} \in \mathbb{R}^n$$

The “forward” system: $A\vec{x} = \vec{b}$

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{m \times n}, \vec{x} \in \mathbb{R}^n, \vec{b} \in \mathbb{R}^m$$

- The column space, $C(A)$, is the set of right-hand sides with a solution.
- The null space, $N(A)$, is the set of solutions to $A\vec{x} = \vec{0}$.
- For every right-hand side in column space there is a unique solution in the row space, $R(A)$.
- Every solution can be written as a sum of unique sum of the solution from the row space and a vector from the null space.

The “forward” system: $A\vec{x} = \vec{b}$

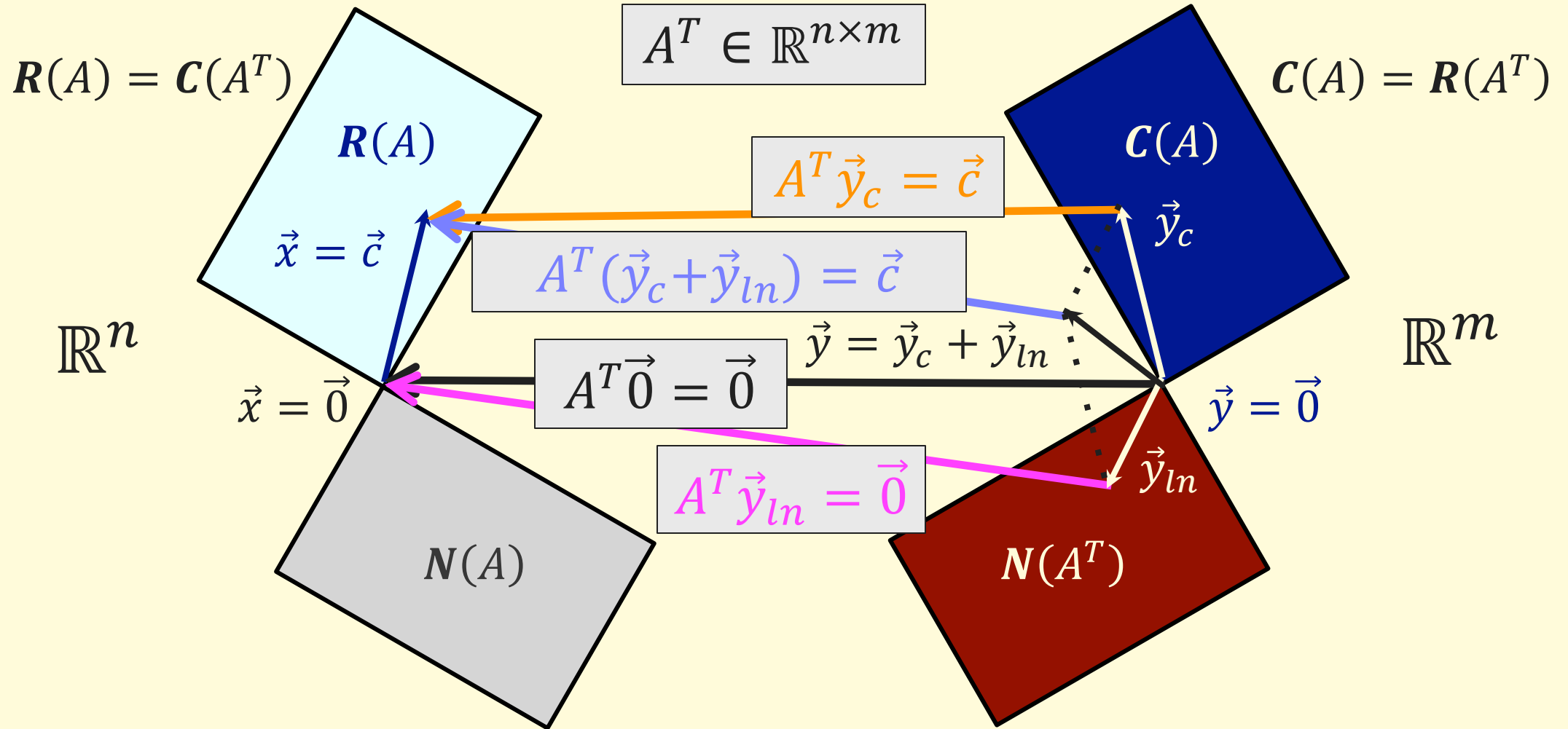


The “backward” system: $A^T \vec{y} = \vec{c}$

$$A^T \vec{y} = \vec{c}, A^T \in \mathbb{R}^{n \times m}, \vec{y} \in \mathbb{R}^m, \vec{c} \in \mathbb{R}^n$$

- The row space, $R(A) = C(A^T)$, is the set of right-hand sides with a solution.
- The left null space, $N(A^T)$, is the set of solutions to $A^T \vec{y} = \vec{0}$.
- For every right-hand side in row space there is a unique solution in the column space, $C(A) = R(A^T)$.
- Every solution can be written as a sum of unique sum of the solution from the column space and a vector from the left null space.

The “backward” system: $A^T \vec{y} = \vec{c}$



Rank and the four subspaces.

Consider a coefficient matrix with m rows, n columns and a rank of r .

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{m \times n}, \text{rank}(A) = r$$

If $r = m = n$:

- **The matrix is full rank (and is square); nonsingular.**
- The column space and row space are both the full parent space.
- The null spaces are both just the origin.
- There is a unique solution to every right-hand side for both $A\vec{x} = \vec{b}$ and $A^T\vec{y} = \vec{c}$.

Rank and the four subspaces.

Consider a coefficient matrix with m rows, n columns and a rank of r .

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{m \times n}, \text{rank}(A) = r$$

If $r = m, m < n$:

- **The matrix is full row rank (and has more columns than rows).**
- The column space is the full parent space, and the left null space is just the origin.
- The row space is not the full parent space, and the null space has dimension $n - r$.
- There are many solutions to every right-hand side for $A\vec{x} = \vec{b}$.
- If $A^T \vec{y} = \vec{c}$ has a solution it will be unique, but not all $\vec{c}$ give a solution.

Rank and the four subspaces.

Consider a coefficient matrix with m rows, n columns and a rank of r .

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{m \times n}, \text{rank}(A) = r$$

If $r = n, m > n$:

- **The matrix is full column rank (and has more rows than columns).**
- The row space is the full parent space, and the null space is just the origin.
- The column space is not the full parent space, and the left null space has dimension $m - r$.
- If $A\vec{x} = \vec{b}$ has a solution it will be unique, but not all $\vec{b}$ give a solution.
- There are many solutions to every right-hand side for $A^T \vec{y} = \vec{c}$.

Rank and the four subspaces.

Consider a coefficient matrix with m rows, n columns and a rank of r .

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{m \times n}, \text{rank}(A) = r$$

If $r < m, r < n$:

- **The matrix is neither full row nor full column rank.**
- Neither the row space nor the column space are the full parent spaces.
- The null space has dimension $n - r$, and the left null space has dimension $m - r$.
- For both $A\vec{x} = \vec{b}$ and $A^T\vec{y} = \vec{c}$, not all right-hand sides give a solution but those that do give many solutions.

Unit 7.2.1

**Worked examples on finding
basis and dimension.**

Row space algorithm

- Find a basis for the linear span of the following vectors, such that the basis vectors have as many zeros as possible.
- What is the dimension of the space?
- What is the parent space?

$$V = \text{span} \left(\left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \\ -1 \end{bmatrix} \right\} \right)$$

Row space algorithm

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 2 \\ -1 \end{bmatrix} \right\}$$

- Create a matrix whose rows are the vectors, then carry out Gaussian elimination to reduced row echelon form

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 \\ 2 & 2 & 1 & 1 \\ 3 & 0 & 1 & 0 \\ 1 & -2 & 2 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 1 \\ 0 & -2 & -3 & -1 \\ 0 & -6 & -5 & -3 \\ 0 & -4 & 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 1 \\ 0 & -2 & -3 & -1 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 6 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 2 & 1 \\ 0 & -2 & -3 & -1 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

Row space algorithm

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 2 \\ -1 \end{bmatrix} \right\}$$

- The non-zero rows are a basis; the number of nonzero rows is the dimension; the parent space is defined by the number of elements in each vector.

$$R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$\dim(\text{span}(T)) = 3 \quad V = \text{span}(T) \subseteq \mathbb{R}^4$$

Casting-out algorithm

- Find a basis for the linear span of the following vectors, such that the basis is a subset of the original set.
- What is the dimension of the space?
- What is the parent space?

$$V = \text{span} \left(\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\} \right)$$

Casting-out algorithm

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

- Create a matrix whose columns are the vectors, then carry out Gaussian elimination to echelon form:

$$A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

Casting-out algorithm

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

- The pivot columns of the original matrix are a basis; the number of pivots is the dimension; the parent space is defined by the number of elements in each vector.

$$A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 2 & 0 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow T = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \right\}$$

$$\dim(\text{span}(T)) = 2 \quad V = \text{span}(T) \subseteq \mathbb{R}^3$$

The four matrix subspaces.

- Find the dimension and a basis for the four fundamental subspaces of the following matrix:

$$A = \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 2 & -2 & 0 & -2 & 2 \\ -1 & 1 & 1 & 2 & 0 \end{bmatrix}$$

The four matrix subspaces.

- Begin with Gaussian elimination to reduce row echelon form:

$$A = \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 2 & -2 & 0 & -2 & 2 \\ -1 & 1 & 1 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 0 & 0 & -4 & -4 & -4 \\ 0 & 0 & 3 & 3 & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 0 & 0 & -4 & -4 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = R$$

The four matrix subspaces.

- The non-zero rows of an echelon form are a basis for the row space:

$$R = \begin{bmatrix} 1 & -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R(A) = \text{span} \left(\begin{bmatrix} 1 \\ -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right), \quad \dim(R(A)) = 2, \quad R(A) \subseteq \mathbb{R}^5$$

The four matrix subspaces.

- The pivot columns of the original matrix are a basis for the column space:

$$R = \begin{bmatrix} 1 & -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, A = \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 2 & -2 & 0 & -2 & 2 \\ -1 & 1 & 1 & 2 & 0 \end{bmatrix}$$

$$\mathcal{C}(A) = \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right), \quad \dim(\mathcal{C}(A)) = 2, \quad \mathcal{C}(A) \subseteq \mathbb{R}^3$$

The four matrix subspaces.

- The special solutions to $R\vec{x} = \vec{0}$ are a basis for the null space:

$$R = \begin{bmatrix} 1 & -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \vec{x}_{n1} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \vec{x}_{n3} = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$N(A) = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right), \quad \dim(N(A)) = 3, \quad N(A) \subseteq \mathbb{R}^5$$

The four matrix subspaces.

- The special solutions to $A^T \vec{y} = \vec{0}$ are a basis for the left null space:

$$A = \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 2 & -2 & 0 & -2 & 2 \\ -1 & 1 & 1 & 2 & 0 \end{bmatrix}, A^T = \begin{bmatrix} 1 & 2 & -1 \\ -1 & -2 & 1 \\ 2 & 0 & 1 \\ 1 & -2 & 2 \\ 3 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & -4 & 3 \\ 0 & -4 & 3 \\ 0 & -4 & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & -1 \\ 0 & -4 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -3/4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & -3/4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = R$$

The four matrix subspaces.

- The special solutions to $A^T \vec{y} = \vec{0}$ are a basis for the null space:

$$R = \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & -3/4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \vec{x}_{ln1} = \begin{bmatrix} -1/2 \\ 3/4 \\ 1 \end{bmatrix}$$

$$N(A^T) = \text{span} \left(\begin{bmatrix} -1/2 \\ 3/4 \\ 1 \end{bmatrix} \right), \quad \dim(N(A^T)) = 1, \quad N(A^T) \subseteq \mathbb{R}^3$$

The four matrix subspaces.

- Double check that the dimensions all match up properly.

$$\dim(\mathbf{R}(A)) = 2, \quad \mathbf{R}(A) \subseteq \mathbb{R}^5$$

$$\dim(\mathbf{N}(A)) = 3, \quad \mathbf{N}(A) \subseteq \mathbb{R}^5$$

$$\dim(\mathbf{R}(A)) + \dim(\mathbf{N}(A)) = 5$$

$$\dim(\mathbf{C}(A)) = 2, \quad \mathbf{C}(A) \subseteq \mathbb{R}^3$$

$$\dim(\mathbf{N}(A^T)) = 1, \quad \mathbf{N}(A^T) \subseteq \mathbb{R}^3$$

$$\dim(\mathbf{C}(A)) + \dim(\mathbf{N}(A^T)) = 3$$